

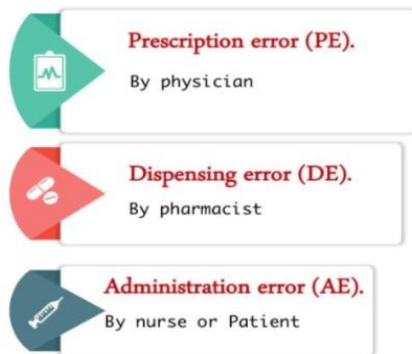


Medication Errors

Definition: Any preventable event that may cause or lead to inappropriate medicine use or patient harm, while the medication is in the control of the health care professional, patient, or consumer"

- IF medical Error reached Patient and cause harm called Adverse Event
- IF medical Error does NOT Reaches Patient called a near miss

Types of Medication Error



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• Classification of medication errors

		Error	Harm	Notes
No error	A	✗	✗	Possibility of error but no error occurred.
Error + No harm	B	✓	✗	Error did not reach patient
	C	✓	✗	Error reach patient → cause no harm
	D	✓	✗	Error reach patient → cause no harm, but need ↑ Monitoring
Error + Harm	E	✓	Temporary harm	Treatment and intervention
	F	✓	Temporary harm	Prolonged hospitalization
	G	✓	Permanent harm	
	H	✓	Near death harm	Anaphylaxis and cardiac arrest
Error + Death	I	✓	Death	

Medication reconciliation

- The process of comparing a patient's medication orders to all of the medications that the patient has been taking.
- This reconciliation is done to avoid medication errors such as omissions, duplications, dosing errors, or drug interactions.
- It should be done at every transition of care in which new medications are ordered or existing orders are rewritten

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A patient is admitted to the hospital for the treatment of atrial fibrillation. During hospital stay, the patient receives IV diltiazem was discharge on oral diltiazem plus his previous antihypertensive medication verapamil. Which of the following medication error-reduction programs would most likely have helped to prevent this duplication of therapy?

- A) Bar Coding
- B) High-Risk Drug Protocol
- C) Medication Reconciliation
- D) Computerized physician order e

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TYPE OF MEDICATION ERRORS

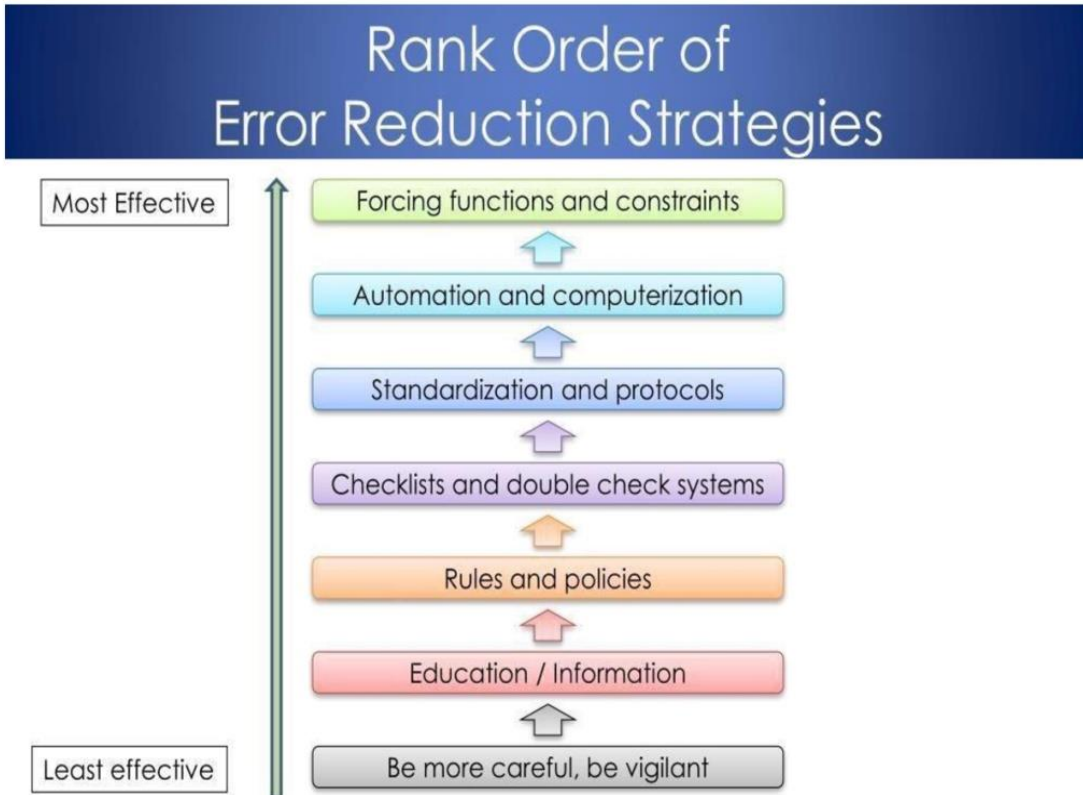
Prescribing	Inappropriate drug selection, dose, dosage form or route of administration.	الدكتور غلط في اختيار الدواء _جرعته_طريقة أخذه
Omission	- Patient doesn't receive ascheduled dose of medication. 2nd most common error in the medication use process.	الدواء مثلا 3 مرات يوميا _ هو خد مرتين (نسي جرعه)
Unauthorizeddrug	- Drug was not authorized by an appropriate prescriber. - Medication given to wrong patient.	المريض اخذ دواء غير الي في الروشته او من غير روشته خالص الممرضة ادت دوا لمريض ثاني _ عكست دوا للمرضين
Wrong time	- Drug not administered in accordance to the interval 1st common error in the medication use process.	رقم 1 في الاخطاء
Wrongdose	The dose administered is different from that prescribed.	خد جرعه غلط غير الي في الروشته
Wrongdosageform	Dosage form different from that prescribed.	مكتوبله نيروتون حقن وانت اديتله نيروتون اقراص
Wrongpreparation	- Reconstitution error - IV admixture compoundin gerror	المريض غلط وهو يبخل المضاد الحيوي
Wrongadministration	- Drug is given to patient inappropriately - IM drug given IV	الدواء اتاخذ غلط هو عضل مثلا وانت اديته وريد
Deteriorated drug	Patient given expired or deteriorated drug	مريض بياخذ دوا بايظ أو تاريخ صالحيته منتهي
Monitoring	Patients are not monitored appropriately	
Compliance	- Patients use medications in an inappropriate way - Patient use metered dose inhaler in wrong way	المريض مش عارف يستخدم الدوا مثال : المريض مش عارف ستخدم بخاخة الصدر
Duplication error	2 drugs from the same class are prescribed.	دوايين من نفس العليه الدوائيه مكتوبين في نفس الروشته

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How To Prevent ME

- 5 `rights' of medication:
 - right patient
 - right drug
 - right dose
 - right route
 - right frequency
- Classification of adverse drug reactions





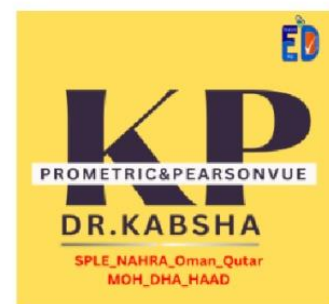
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• **Classification of adverse drug reactions**

	type	Feature	Examples	Management
A	Augmented Related to pharmacological Dose related	• Common • Predictable • Low mortality (↓↓)	1. Drug with narrow TI (dose related) 2. Toxic effects of drugs: • Digoxin toxicity • Serotonin toxicity with SSRI • Anticholinergic effects of TCA	Reduce dose or withhold
B	Bizarre Not related to pharmacological action Non-dose related	• Un common • Not related to pharmacological action • Unpredictable • High mortality (↑↑) • Non-dose related	1. Penicillin hypersensitivity (immune reaction) 2. Idiosyncratic reactions: • Acute porphyria • Malignant hyperthermia • Pseudo allergy	withhold and avoid in future
C	Chronic Dose related Time related	• Un common • Dose related → Related to cumulative dose • Time related	1. Cortisone for long time cause suppression of hypothalamic pituitary adrenal axis.	✓ Reduce dose or withhold ✓ Withdrawal may have to be prolonged
D	Delayed Time related	• Uncommon • Time related	1. Teratogenicity 2. Carcinogenesis 3. tardive	✓ intractable
E	End of use (withdrawal)	• Uncommon • Occur soon after withdrawal of drug.	1. Opiate withdrawal syndrome 2. Beta blocker withdrawal • Myocardial ischemia	✓ Reintroduced ✓ Withdraw slowly
F	Failure	• Common • Often caused by drug interaction	1. Oral contraceptive failure when taken with enzyme inducer drug	✓ Increase dosage

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Pharmacovigilance (PV)	
Definition	Pharmacovigilance (PV) is defined as the science and activities relating to the detection, assessment, understanding and prevention of adverse effects or any other drug-related problem.
Aim	The aims of PV are to enhance patient care and patient safety in relation to the use of medicines; and to support public health programmers by providing reliable, balanced information for the effective assessment of the risk-benefit profile of medicines.
Difference between Active and Passive Pharmacovigilance (PV)	
Active (PV)	- Active (or proactive) safety surveillance means that active measures are taken to detect adverse events. - This is managed by active follow-up after treatment and the events may be detected by asking patients directly or screening patient records. This surveillance is best done prospectively. - Active pharmacovigilance is sometimes very descriptively referred to as hot pursuit. - The most comprehensive method is cohort event monitoring (CEM). - Cohort event monitoring is often referred to as prescription event monitoring (PEM), but this terminology is inappropriate where individual prescriptions with subsequent dispensing are not part of the process of provision of medicines.
Passive (PV)	- Passive surveillance means that no active measures are taken to look for adverse effects other than the encouragement of health professionals and others to report safety concerns. - Reporting is entirely dependent on the initiative and motivation of the potential reporters. This is the most common form of pharmacovigilance. - It is commonly referred to as “spontaneous” or “voluntary” reporting. - In some countries this form of reporting is mandatory.



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Primary Resources	
Definition	- Primary resources contain first-hand information, meaning that you are reading the author's own account on a specific topic or event that s/he participated in. Review articles or editorials are not primary literature.
Examples	Journal articles of original research, conference papers, dissertations, technical reports, patents. - Several types of publications considered primary, including controlled trials, cohort studies, case series, and case reports.
Adv	- Access to detailed information about a topic - The ability to personally assess the utility and validity of study results. - Tends to be more recent than tertiary or secondary literature.
Dis Adv	Misleading conclusions based on only one trial without the context of other researches, The need to have good skills in medical literature evaluation, - The time needed to evaluate the large volume of literature available.
Secondary resources	
Definition	Secondary resources include indexing and abstracting systems that organize and provide easy retrieval of primary resources.
Examples	- MEDLINE (through PubMed, EBSCO, Ovid), - Academic Search Premier, - Cochrane Database of Systematic Reviews, - Iowa Drug Information Service (IDIS), - International Pharmaceutical Abstracts (IPA), - Embase/Excerpta Medica, - Biosis Previews/Biological Abstracts, - CancerLit, SedBase, Reactions, - Clin-Alert, Current Contents, and Toxline. Proper training
Tertiary Resources	
Definition	Tertiary sources consist of primary and secondary source information which has been collected and distilled. They present summaries of or an introduction to the current state of research on a topic, summarize or condense information from primary and secondary sources
Examples	electronic databases (e.g., Micromedex, Lexicomp) review articles. Facts and Comparisons Uptodate
Dis Adv.	- The lag time associated with publication, resulting in less current information - The information in a tertiary text may be incomplete due either to : - Space limitations of the book or - Incomplete literature searches by the author - Errors in transcription, human bias, incorrect interpretation of information, or a lack of expertise by authors.

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Clinical trials	Pubmed
Pharmaceutical compounding	• Trissel's stability of compounded formulations • Remington the science and practice of pharmacy
International drugs	Martindale the complete drug reference
Pediatric dosage recommendation.	The Harriet Lane Handbook
Drug-Drug interactions.	USP DI Vol.I
Drug-food interactions.	USP DI Vol.II
Adverse drug reaction	Meyler

• **Color drug references**

Orange book provides a list of approved drug products

•

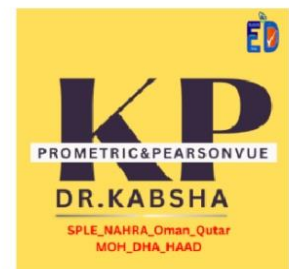
Pink book: epidemiology and vaccine-preventable diseases

Purple boo biological drug products

Green book animal drug products

Yellow book health information for healthcare professionals

Red book drug pricing information



2- Which of the following could be considered as primary literature

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- A. ☒ Letters to the editor
- B. clinical trials
- C. Meta-Analysis
- D. Clinical guidelines

3- Which of the following statements is TRUE ?

- A. Prospective studies evaluate events in the past .
- B. ☒ Retrospective studies are useful for investigating rare diseases .
- C. A run-in phase typically increases variation .
- D. Crossover design allows patients to administer only one type of treatment

4- Which of the following is not important when evaluating a clinical study ?

- A. Study objective
- B. Patients' inclusion and exclusion criteria
- C. Doses of drugs administered
- D. ☒ Number of authors

5- Which of the following is considered inappropriate while developing an effective search strategy ?

- A. Determine if the question is research related or clinical
- B. Identify appropriate keywords
- C. ☒ Focusing only on the medication included in the requester inquiry
- D. Determine the Profession of the inquirer

6- Which of the following statements is TRUE about tertiary resources?

- A. Tertiary resources should not include a bibliography .
- B. Textbooks usually include the most recent literature and/or information .
- C. ☒ Tertiary resources include computer databases and textbooks
- D. The credentials and expertise of the authors of a tertiary resource are not important .

7- Which of the following resources would be appropriate for identifying a drug manufactured in a foreign country ?

- A. ☒ Martindale: The Complete Drug Reference
- B. Clinical Pharmacology
- C. Trissel's Stability of Compounded Formulations
- D. AHFS

8- Which of the following resources could be appropriate for identifying the tablets with the imprint code of APO-

K 600 ?

- A. Identidex
- B. Lexi-Drugs ID (Lexicomp Online)
- C. Briggs
- D. ☒ ☒ A and B

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9- Which of the following resources could be appropriate for determining the adverse effects of ginkgo biloba ?

- A. Altmedex
- B. Natural medicines database
- C. Trissel
- D. A and B

10- Which of the following resources would be appropriate in evaluating the drug interaction between Clopidogrel and escitalopram?

- A. Lexi-Interact
- B. Trissel
- C. Stockley
- D. A and C

11- Which of the following resources would be appropriate for determining whether Amlodipine causes edema?

- A. Briggs
- B. Trissel
- C. Myler
- D. Stockley

12- What resource should be used when searching for the most current clinical trials on the use of novel anticoagulants for pulmonary embolism?

- A. Pharmacotherapy, the pathophysiologic approach
- B. PubMed
- C. Uptodate
- D. Micromedex

❖ **Medication safety in pregnancy and lactation**

- Hale's medications and mothers' milk
- Brigg's drugs in pregnancy and lactation

❖ **Drug interactions**

- Stockley's drug interactions
- Lexi interact

❖ **Toxicology**

- Gold frank's toxicological emergencies
- Casarett and Doull's toxicology

IV medications compatibility and stability

- ASHP's interactive handbook on injectable drugs

Best source for off label use?

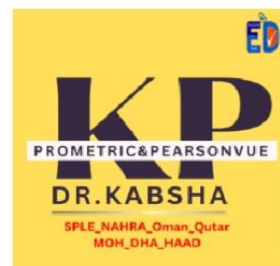
- A. AHFS Drug Information.
- B. Daily Med.
- C. Drugs@FDA.
- D. Manufacturer-provided PI

Which of the following resources is best to use for common compounding techniques and ingredients?

- A. Red book
- B. Drug facts and comparison
- C. Martindale: the complete drug reference
- D. Remington: the science and practice of pharmacy

The pharmacy received a call from the Intensive Care Unit nurse who has a patient receiving an intravenous solution containing amoxicillin. The doctor has now ordered morphine to be injected into the Y site of the intravenous solution every 4 hours. The nurse wants to know if these drugs will be compatible at the Y site. From which references the pharmacist should provide the required information?

- A. Red book
- B. Merck manual
- C. Review of natural products
- D. Handbook on injectable drugs



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Usp 795 —————→ non sterile preparation

USP 797 —————→ sterile preparation

USP 800 —————→ hazardous drug

- **Vertical** Flow Hoods: **Chemotherapy's** medication
 - **Horizontal** Flow Hoods: Non Chemotherapy's - Air Exhausted grille
- Low risk: Single sterile dosage using aseptic technique within an ISO Class 5 using sterile needles
- Medium risk: multiple sterile dosages x multiple doses of sterile products total parenteral nutrition).
- High risk: non sterile Prepared from nonsterile ingredients
Compounded using a nonsterile device prior to terminal sterilization
Containing nonsterile water that are stored for more than 6 hours before sterilization
- Terminal sterilization: cleaning at end of work - Thermos table produces
 - Aseptic sterilization: The pharmacist use it in IV room as always – Thermo liable

INTERNATIONAL ORGANIZATION OF STANDARDIZATION (ISO) CLASSIFICATION OF PARTICULATE MATTER IN ROOM AIR	
ISO Class	ISO (Particles per cubic meter)
3	35.2
4	352
5	3,520
6	35,200
7	352,000
8	3,520,000



**Comparison between selected equivalents
of FS 209 and ISO 14644-1**

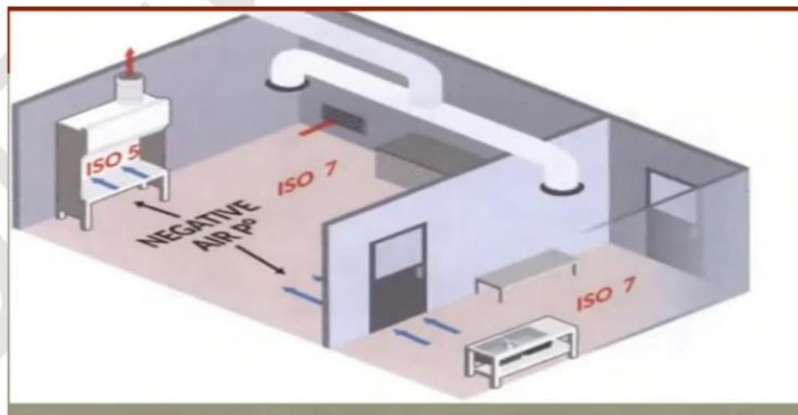
ISO 14644-1	FS 209
Classes	
ISO 3	1
ISO 4	10
ISO 5	100
ISO 6	1,000
ISO 7	10,000
ISO 8	100,000

each individual involved in manufacturing of sterile preparations is required to be well trained to produce sterile preparations, which is the following tests is required by USP to be done annually to ensure
Consistent production of sterile drugs:

A-sterility test

B-media fill

C-viable and non-viable



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PATIENT ADHERENCE

Medication adherence is defined as the extent to which a patient's medication taking behavior coincides with the intention of the health advices given.

COST OF NON-ADHERENCE

Studies showed that the treatment costs increases due to non-adherence because of the following:

- ❖ Increased clinical visits.
- ❖ Larger number of medications for the same patient.
- ❖ Increased dosage changes.
- ❖ Increased hospital admission.
- ❖ Increased duration of stay in the hospital.
- ❖ It was reported that poor adherence with post immune suppression regimens is a leading cause of rejected organ transplant.

CATEGORIES OF MEDICATION NON- ADHERENCE

Intentional

- ❖ The patient does not adhere to the prescription intentionally.
- ❖ The patient may take less prescribed dose with the assumption that the prescribed dose is higher.
- ❖ The patient may take more than prescribed to get quicker relief or recovery.

Unintentional

- ❖ Happens when patient misunderstood or forgotten the prescriber's or pharmacist directions.
- ❖ Fails to adhere due to cognitive problems

METHODS TO DETECT MEDICATION NON ADHERENCE

1. Direct- Objective :

- ❖ Measure blood or urine levels of drug

2. Indirect –Objective

- ❖ Pill count- counts the tablets remaining in the container.
- ❖ Prescription refill- accurate data monitoring system.
- ❖ Electronic medication containers-opening and closing times of the container recorded on a microprocessor in the lid of the container.

3. Health outcome measures

- ❖ Assessing therapeutic efficacy.

Eg: blood pressure control, asthma severity, survival etc

4. Utilization of healthcare services

- ❖ Clinical attendance.
- ❖ Appointment making.
- ❖ Appointment keeping.
- ❖ Preventive visits.

5. Indirect- subjective

- ❖ Patient interview.
- ❖ Diary keeping

DETERMINANTS OF MEDICATION ADHERENCE

1.Predisposing

- ❖ Demographic factors.
- ❖ Patient knowledge, attitudes, beliefs and perception about illness and its severity cause, prevention and treatment.

2. Enabling factors

- ❖ Skills and resources needed for adherence.

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- ❖ Skills refer to the patient's ability to adopt behaviors which will assess adherence.
- ❖ Resources include the availability and accessibility of healthcare facilities

3. Reinforcement factors

- ❖ Support by family members, healthcare providers, local community and society in general.
- ❖ Factors associated with the treating physician such as communication with the patient, time spent with the patient etc
- ❖ Giving written instructions to the patient.
- ❖ Social isolation associated with diseases such as tuberculosis.

4. Factors associated with medication itself.

- ❖ Long duration of treatment and adverse effects may lead to poor adherence.
- ❖ The complexity of the regimen

Which of the following is used as an electronic medication dispenser to improve medication adherence?

- A) Text messaging
- B) Cell phone calendar
- C) E-pill MultiAlarm TimeCap
- D) Med-E-Lert Automatic pill Dispenser

Essential amino acids

I	Isoleucine
Love	Lysine
Lucy	Leucine
Very	Valine
Much	Methionine
Please	Phenylalanine
Try	Tryptophan
To	Threonine
Help	Histidine
Arginine	Arginine

- M A L C cysteine (**acidic**), Lysine, Arginine and Methionine. →→ Present in hair and nail. Produce keratine
- BCAA amino acids(**leucine** , isoleucine, valine) decrease **depression & anxiety**

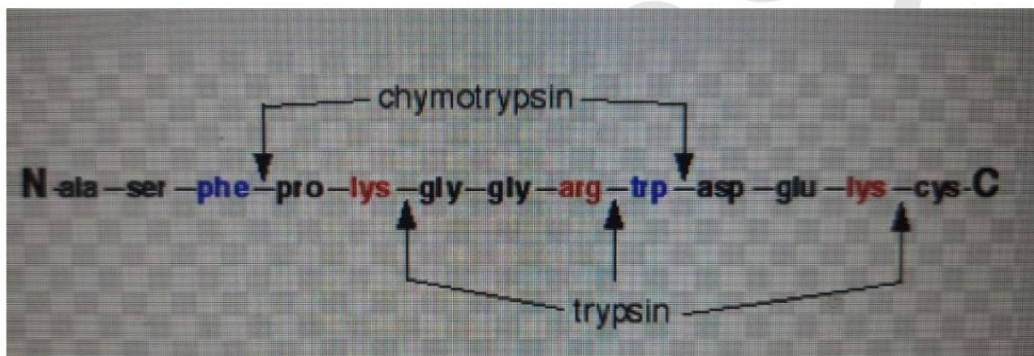
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The enzyme pepsin in digestion breaks down any part of the protein

- A. Serine
- B. Lysie
- C. Phenyl alanine
- D. Leucine

→ Pepsin on Cyclic aminoacids tryptophan phenylalanine tyrosine



Which c terminus amino acid does trypsin cleave?

- A. leucine
- B. methionine
- C. phenylalanine
- D. lysine

→ **Monosaccharides:** Glucose Galactose Fructose

→ **Disaccharides :** Sucrose Maltose Lactose

→ **Polysaccharides :** Starch Glycogen Cellulose

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Calories

1g fat = 9 calories

1g protein = 4 calories

1g carb = 4 calories

10 g of fat =???

1 g dextrose kcal:

- A. 3
- B. 3.4✓
- C. 5
- D. 34

Aspirin to prevent colorectal cancer 300 mg for 5 years
Fit &CEA

What is Casualty score? Narnjo
No renal adjustment • DPP 4 inhibitor: Linagliptin.

Vancomycin
Ca correction?
Phenytoin

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Verbal communication
Verbal communication is communication involving words, both spoken and written. Learn how to maximize the effectiveness of the words you choose
non-verbal communication
This is a complete mix of behaviors, psychological responses, and environmental interactions through which we consciously or unconsciously communicate with another person.
Main elements of non-verbal communication
▶ Personal appearance of communicator(s) ▶ Environment of the encounter ▶ Proxemics : Most pharmacist-patient interactions take place within the sociable zone (1-3m). ▶ Body movements (Kinesics) Avoid: Swinging foot & tapping fingers Avoid Gesture like crossing the legs, Avoid biting nails ,Avoid shifting eyes & head quickly during conversation ▶ Paralanguage: is the quality of sound and manner of delivery that accompany the spoken words. This can be more reliable than the words themselves ▶ Touch , Facial expressions , Eye behavior and Postures
Open-ended questions:
Open ended questions are that start with who, what, where, when, how and why and require more than a yes/no
Closed-ended questions:
<i>Closed ended questions and leading/restrictive questions gives Yes/No response</i>
Passive communication
<i>Passive communication involves violating your own rights by failing to express honest feelings, thoughts, and beliefs</i>
Aggressive Communication
<i>It is a style in which individuals express their feelings and opinions in a way that violates the rights of others</i>
Passive-Aggressive communication
<i>It involves expressing your needs and feelings in an unclear and confusing manner.</i>
Assertive Communication
<i>Assertiveness is the ability to state positively and constructively your rights or needs without violating the rights of others</i>
Patient-Oriented Process Skills
1. <u>Knock on the door and request permission to enter the room of the institutionalized patient.</u> 2. <u>Introduce yourself.</u> 3. <u>Try to achieve privacy.</u> 4. <u>Make sure the patient is comfortable.</u> 5. <u>Communicate at eye level or lower.</u> 6. <u>Remove distractions (loud television, radio, relatives, and friends).</u> 7. <u>Clarify the purpose of the interview.</u> 8. <u>Obtain the patient's permission for the interview.</u> 9. <u>Verify the patient's name and correct pronunciation.</u> 10. <u>Address the patient by the appropriate title.</u> 11. <u>Maintain eye contact with the patient.</u>

Which of the following communication techniques is the most appropriate at the message you want to deliver to a patient (or healthcare provider) is received correctly?

- A. Feedback
- B. Direct contact
- C. High voice tone
- D. Non-verbal communication

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BATHE technique:

A psychotherapeutic procedure, meaning it seeks to empower patients to trust themselves and others, confirm their positive feelings about themselves, and enhance their ability to control the circumstances of their lives.

Used in severe situations (e.g., when the patient is in extreme pain, when the patient is psychotic anxiety, depression, and situational stress disorders or suicidal).

Problems that have an emotional component or that affect quality of life. For example, identify how a patient is coping with a new diagnosis



B. Background: Use an open-ended question similar to the standard question to elicit the chief concern.

A Affecting: Solicit feelings or effect on quality of life. “How do you feel about this?” or “How is this problem affecting your life?”

T. Troubling: Identify the relative importance or specific areas of concern. “What troubles you the most about...?”

H. Handling: “How are you dealing/coping with this problem?”

E Empathy: Reflect back the concern and/or emotion. “You seem frustrated by the lack of...”

COUNCEKING ?

Be non-judgmental

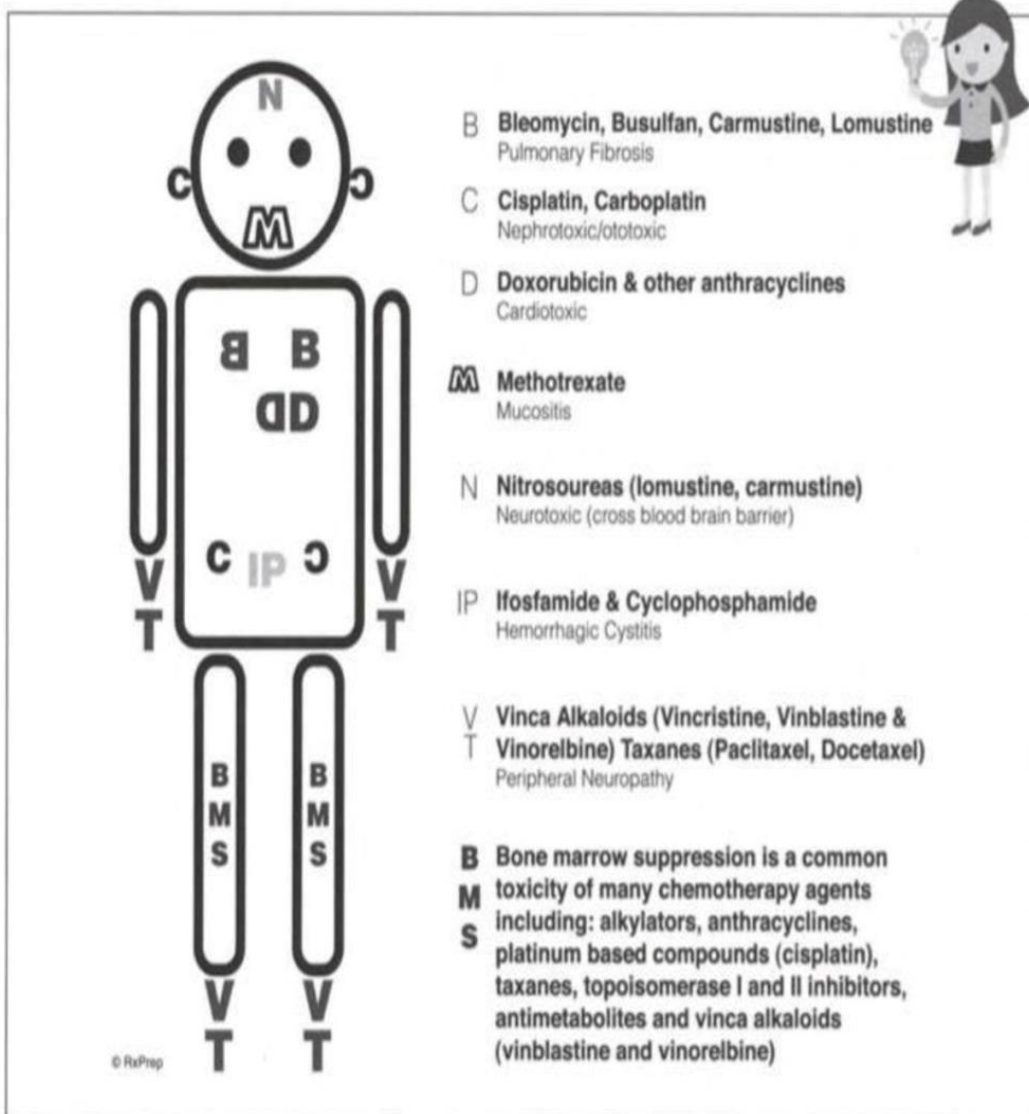
Active listener

Uncross arms and legs

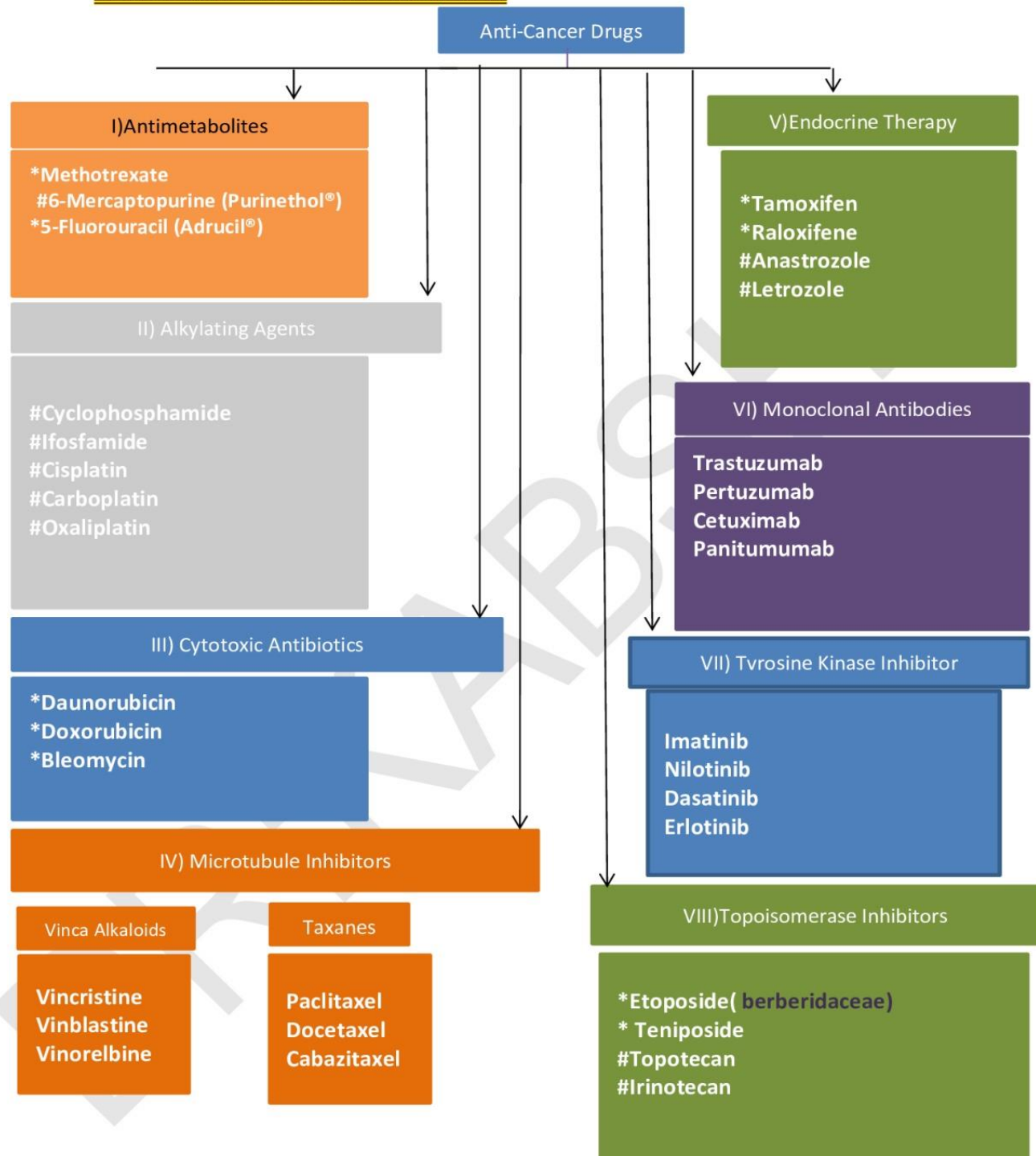
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Prevalent Toxicities of Common Chemotherapy Agents

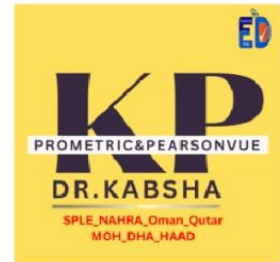
This is a visual way to remember the toxicities associated with common chemotherapy agents. Students are encouraged to draw a chemo person to help make connections and learn the material.



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• **Breast cancer**

- Antiestrogen USE in breast cancer: **tamoxifen**
 - Immunosuppressant cause hirsutism: **cyclosporine**
- Irinotican (topoisomerase inhibitor) in **breast cancer**

Which of the following anti-cancer alkaloids act by stabilizing the microtubule polymer and protecting it from disassembly?

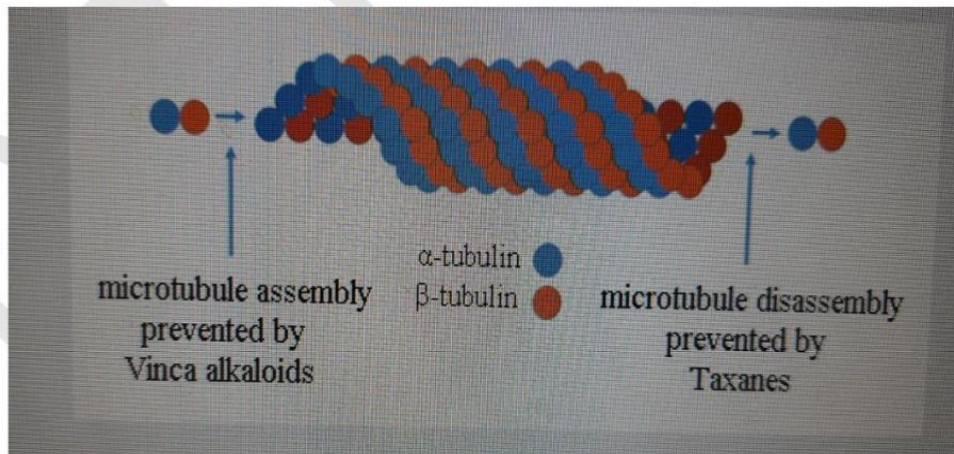
- A. Paclitaxel
- B. Colchicine
- C. Vinblastine
- D. Camptothecin

Which of the following drugs is associated with progressive multifocal leukoencephalopathy side effect?

- A. Bisoprolol
- B. Pregabalin
- C. Doxycycline
- D. Natalizumab

Which of the following medications is an aromatase inhibitor used. For the treatment of breast cancer?

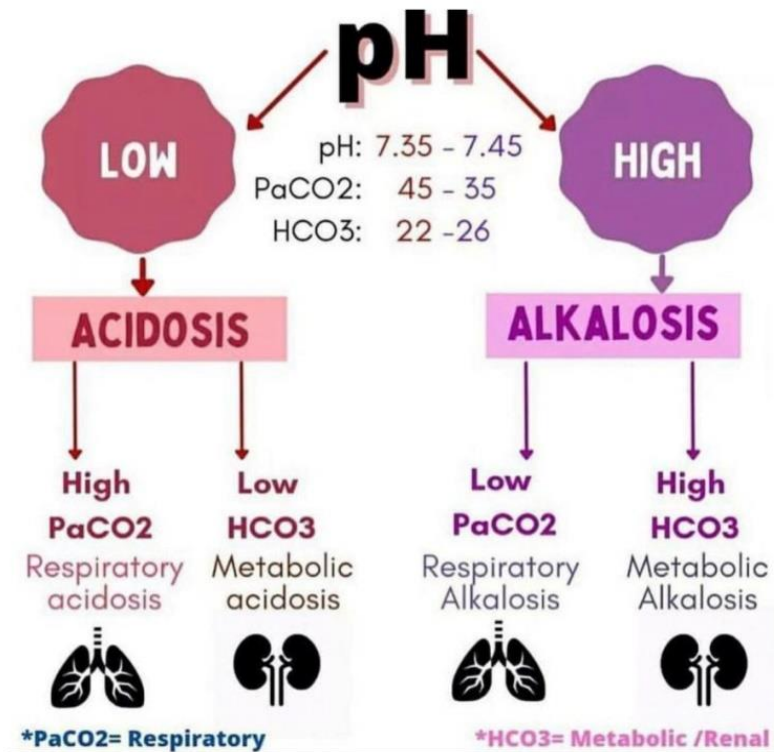
- A. flutamide
- B. octreotide
- C. tamoxifen
- D. anastrozole



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ISOTONIC	HYPOTONIC	HYPERTONIC
• 0.9% Nacl • Lactated Ringer • Ringers' Solution • 5% Dextrose in water	• 0.45% Nacl • 0.33% Nacl • 0.2 % Nacl • 2.5% Dextrose water	• 3% Nacl • 5% Nacl • 3%Nacl or 5% Nacl +D/W • >5% D/W example, D10W
No effect on cell	Swelling	Shrink
3% Nacl ...in central line		

لا يسمح بتداول أي جهد خارج
المشاركين في الكورس وتعتبر أمانه



- Ttt of hypophosphatemia: sevelamer, ca acetate
- Ttt of hypophosphatemia: add Vit D
- Ttt of hyperkalemia: sodium polystyrene
- Ttt of hypokalemia:

- 10 - 20 meq/hour Standard IV replacement rate
- 20 - 40 meq/hour Serum potassium < 2.5 meq/L or moderate-severe symptoms
- > 40 meq/hour Serum potassium < 2.0 Meq/L or life-threatening symptoms



- Premature labor Stop premature labor ♦: **Ritodrine**
- Treatment of preterm labor is "Tocolytic": **MgSO₄, Indomethacin, and Nifedipine**
- Eclampsia: **Mg sulfate**
- **Abortion:**
- Substance that induces abortion: **Misoprostol, Mifepristone, Methotrexate** Ant estrogen cause abortion: **Anastrozole**
- Ant progesterone cause abortion: **Mifepristone, Misoprostol**
- Prevention and treatment of postpartum and post abortion hemorrhage: **Ergonovine**
- **Oral contraceptive** used to treat: endometriosis-associate pain "pain in menses and during intercourses"
- nfertility in women who do not ovulate: **Clomiphene citrate** ————

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Stop before:

❖ Stop before pregnancy:

1 month: **Isotretinoin** (acne)

2 months: **Neostigmine** (myasthenia gravis), **Fingolimod** (multiple sclerosis)

6 months: **Ribavirin** #

2 years: **Leflunomide** (RA treatment)

Mycophenolate (immunosuppressant) o

6 weeks in female o

12 weeks in male

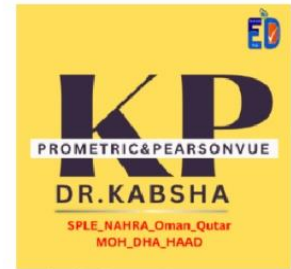
❖ Fungal:

- Pregnant with candida albinos: **Clotrimazole**
- Anti-fungal CI in pregnancy “: **Griseofulvin, Ketoconazole, Voriconazole, Fluty sine,**
- Anti-fungal SAFE in pregnancy: **Amphotericin B, Topical imidazole, NY statin**
- Anti-fungal CI in lactation: **Ketoconazole, Itraconazole,**
- Anti-fungal SAFE in lactation: **Fluconazole**
- Azole that **DOC** for Aspergillus: **VORICONAZOLE**

MOA AZOLE: interfere with the synthesis and permeability of fungal cell membranes by 14-alpha-sterol demethylase. This enzyme is required for the synthesis of ergosterol, the major sterol of most fungal cell membranes

Tinea capitis: fungal infection **scalp & hair**

First line treatment for esophageal candidiasis?



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❖ Hepatic:

- Hepatotoxicity of paracetamol due to: **glutathione depletion**
- Lactulose in Hepatic Encephalopathy (HE): **to decrease ammonia level in blood**
- Hb1 genotype hepatic pt. (HCV) ttt: **Interferon, Ribavirin**
- Fat soluble Vit. **Important for liver: Vit (A.D.E.K)**
- Precursor of bile: **cholesterol**
- Vitamins involved in fatty acid synthesis: **B5, B3, B2 # pancreatic patient # Pancreatic lipase more specific than amylase**
- Neuromuscular blocker used in renal or hepatic failure: **Atracurim**

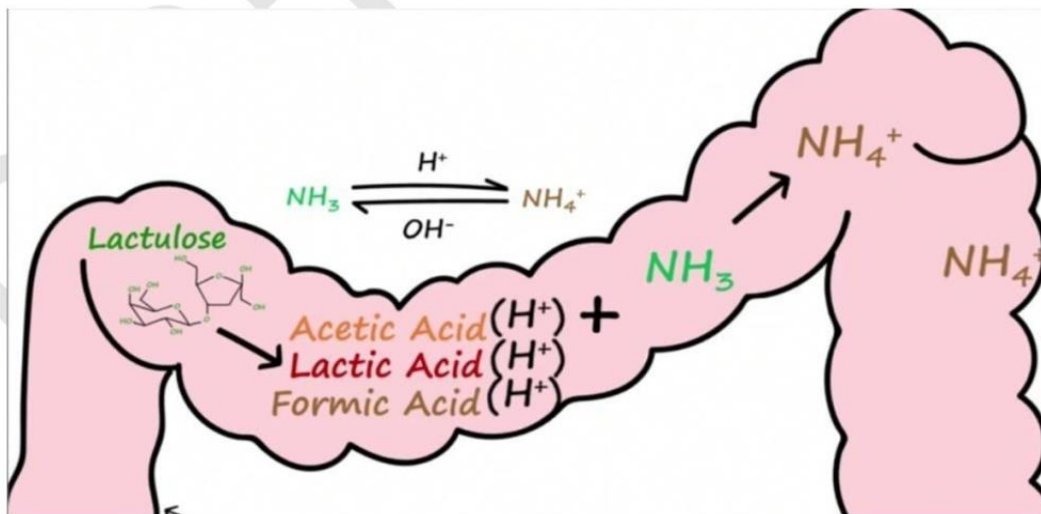
High amylase: **pancreatitis**

High liver enzyme $\uparrow$ (AST & ALT) $\downarrow$ low albumin level: **cirrhosis**

High liver enzyme $\uparrow$ (AST & ALT) $\uparrow$ GGT $\uparrow$ Bilirubin: **Cholestasis**

Cholestasis: Flow of bile Decreased or Blocked

(Steatosis): Fatty liver



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Preparation راجع ملف herbal

Prepare emulsion: **continental**, dry gum method, wet gum method

Not consider formation of suspension: use chelating agent

Boric acid in preparation: **puffer**

Vit c preservative

Gums used in tableting as: **Binding agent**

Aspartame added to preparation as: **sweeting agent**

Bentonite used as: **suspending agent** and Rheological agent

Used to prepare suppository: **Cocoa butter**

Oleaginous base: **white petroleum**

Vegum ?

PEG??

Water: Distilled water use in cold cream:

: sterile water for injection use in large amount of parental

Normal water used for preparation of: **External preparation**

Pka for normal water in room temp: 14

Syndromes:

Toxidrome: The syndrome caused by toxin

Prodrome: the signs and symptoms of toxicity syndrome

Early signs or symptoms

Gray baby syndrome \$: **Chloramphenicol ??**

Gray man syndrome: **Amiodarone**

Red man syndrome: **vancomycin**

Blue-Gray skin discoloration: **amiodarone**

Brown-pink skin discoloration: **Clofazimine**

Purple gloves syndrome ⚡: **phenytoin**

Purple toe syndrome ⚡: **warfarin**

Hand-foot syndrome &: **capecitabine**

Yellow nail syndrome ⚡: **Sodium aurothiomalate**

Lupus like syndrome: **hydralazine & procainamide & isoniazid**

Monday syndrome: **Isosorbide**

Ethambutol :Red-green color blindness:

Reye syndrome: aspirin (children and teenagers) # Not use bismuth in children and teenagers who recovering from flu, chickenpox, viral infection ->

Treatment of **Raynaud's** phenomenon: \$ CCBs: **Nipeditine** \$ VD: **Sildenafil**

Golgi apparatus: # Membrane bound organelle found in mast cell. #

Responsible for packing proteins into vesicles to secretion and therefor plays a key role in the secretory pathway

Remember?????????

Epitope: "Antigenic determinant" Part of an antigen that is recognized by the

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immune system. Specially by antibodies, B cell, T cell

Anergy is a term in immunobiology that describes a lack of reaction by the body's defense mechanisms to foreign substances,

Myasthenia Gravis: Treatment: § Cholinergic: Neostigmine, Physostigmine #
MOA: block the action of Acetylcholinesterase à increasing Acetylcholine level

Multiple Sclerosis (MS): Multiple sclerosis: **Fingolimod** (1st line oral) # DC 2 months before getting pregnant & during pregnancy

Cardiotoxicity????

Remember ????

Cryptococcus neoformans: encapsulated yeast ??? remember bacteria?

Antidote

Activated charcoal give within 4 hours of ingestion

Acetaminophen: Acetylcysteine

Organophosphate, nerve gases (inhibit AChE) so incr Ach :

Atropine(anticholinergic), pralidoxime

Ach: Atropine

Anticholinergic: physostigmine

Pilocarpine: atropine

CCB's: CaCL, Glucagon

Iron: Deferoxamine

Heavy metal: Dimercaprol, DMSA

Lithium: Sodium bicarbonate, Polystyrene Sulfonate

Methotrexate: leucovorin

Heparin: protamine sulfate

Warfarin, anticoagulant: phytonadione "Vit K1"

Dapigatran :

Rivaroxiban

Sulfonylureas: octreotide

Valproic acid: L-Carnitine

Serotonin syndrome: Cyproheptadine, BZDs

Thrombolytics agent: Aminocaproic Acid, Tranexamic Acid

Neuroleptic malignant syndrome, stimulant induce hyperthermia: Dantrolene

MOA of dantrolene: postsynaptic muscle relaxant inhibitors Ca ions release

باقى فى ملف الاسئلة

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Skin conditions:

Psoriasis:

- Coal tar
- Calcipotriol (Vit D)

Acne:

Topical: Benzyl peroxide (Keratolytic) , Azelaic acid

Systemic: Erythromycin, doxycycline ,Isotretinoin

Isotretinoin should be STOP 1 month before getting pregnant

SPF

Woman who has not been in the sun for 4 months develops redness on her chest after lying in the sun for 20 min the next day she applies lotion and develop the same degree of redness in 2hr & 20min ???spf ?

$140/20 = 7$

1st order: linear # rate direct proportional of the conc.

0 order: Non-linear # rate is independent of the conc.

WAATTP, non-linear Warfarin, Alcohol, Aspirin, Theophylline, Tolbutamide, Phenytoin

Largest capsule ◆: 3cm (000)

Med. can be crushed: immediate release tablet ◆

Nasogastric tube ?

Dumping ?

Nausea & Vomiting:

Dopamine (D2) antagonist: Metoclopramide. # Cross BBB & cause EPS

Domperidone # Not cross BBB but cause QT prolongation

Pregnant with N&V: -

Meclizine + Pyridoxine (Vit B6)

- Cyclizine + Pyridoxine (Vit B6) –

Doxylamine + Pyridoxine (Vit B6)

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Rheumatoid Arthritis (RA):

Disease modifying antirheumatic drugs (DMARDs)

A. Non-biologic DMARDs:

1. **Methotrexate:** (1st line) # ??
2. **Leflunomide:** # female should DC 2 years before getting pregnant

B. Biological DMARDs:

1. **TNF-a inhibitors:** Etanercept, infliximab, adalimumab 2.
- 2-**T-cell activation** blockade: Abatacept
3. **B-cell depletion:** Rituximab
4. **IL-6 inhibitors:** Tocilizumab #

Most medication increase the **risk of TB infection** #

Live vaccine should be avoided to avoid the risk of infection Metal use in treatment of RA: Gold ◆ -----

Rheumatoid arthritis take methotrexate and still not improved after 3 months?

- A- add Adalimumab and abatacept
B- add HCQ ✓(Hydroxychloroquine)
C- continue methotrexate



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Gout Treatment:

Acute gout attack:

→ **1st line** in acute is **NSAIDs**: **Indomethacin (DOC)??heart?** , Naproxen, Sulindac
c.i ?

Avoid use in severe renal disease (UA is renally cleared and patients with
(have renal insufficiency) and CVD risk

→ **2nd line: Colchicine: Plant Alkaloid** # Also used in Bechet's disease §

Start **within 36 hours** of symptom onset (for treatment)

CrCl < 30 ml/min: the treatment

dose is the same, but do not give

again for 2 weeks

If risk of myelosuppression, GI and neuromuscular adverse effects in elderly
with CrCl < 30 ml/min; .decrease ,dose and monitor, or use steroid as an alternative

→ **3rd line: Corticosteroids**

Acute side effects of steroids, including 'BG'BP,insomnia, 'appetite

Chronic gout:

Allopurinol (Decrease uric acid synthesis) { Xanthine oxidase inhibitors (XOI) } §

S/E :

Cause severe cutaneous (red skin rash).

Dose :

Allopurinol dose depend on Crcl:

Start at 100 mg daily, then

slowly titrate up until UA is

< 6 mg/dl

doses> 300 mg

can be necessary and should

be divided BID)

CrCl ≤ 30 ml/min: Start at

50 mg daily, increase gradually

up 300 mg/day

Lesinurad #Use only with XOI if UA goals not reached with XOI alone; keep hydrated

Probenecid (Uricosuric)

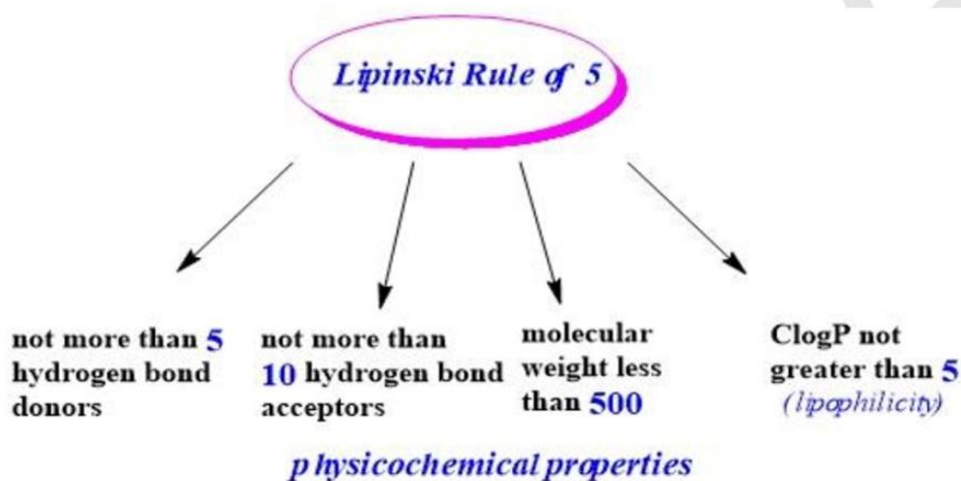
: Increase uric acid excretion which lead to decrease the uric acid Conc.

Drug increase uric ??

Diuretics (Thiazide, Loop diuretics) §

Pyrazinamide, Ethambutol (TB antibiotics) # Both of them increasing the uric acid level

To be orally active ?

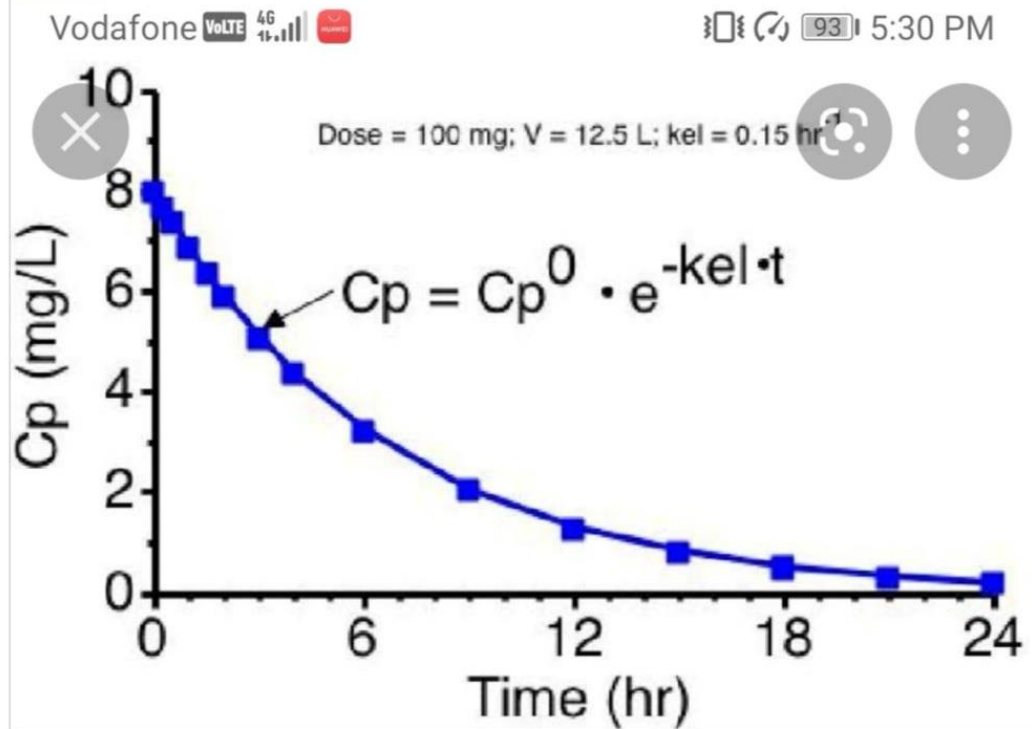


MW < 500 Da
ClogP < 5
H-bond donor < 5
H-bond acceptor < 10

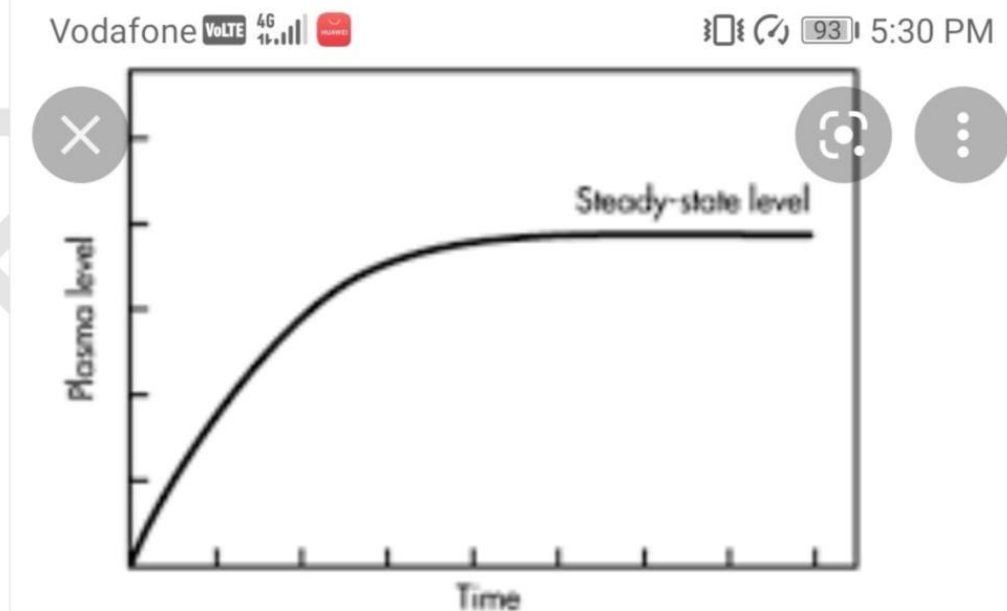


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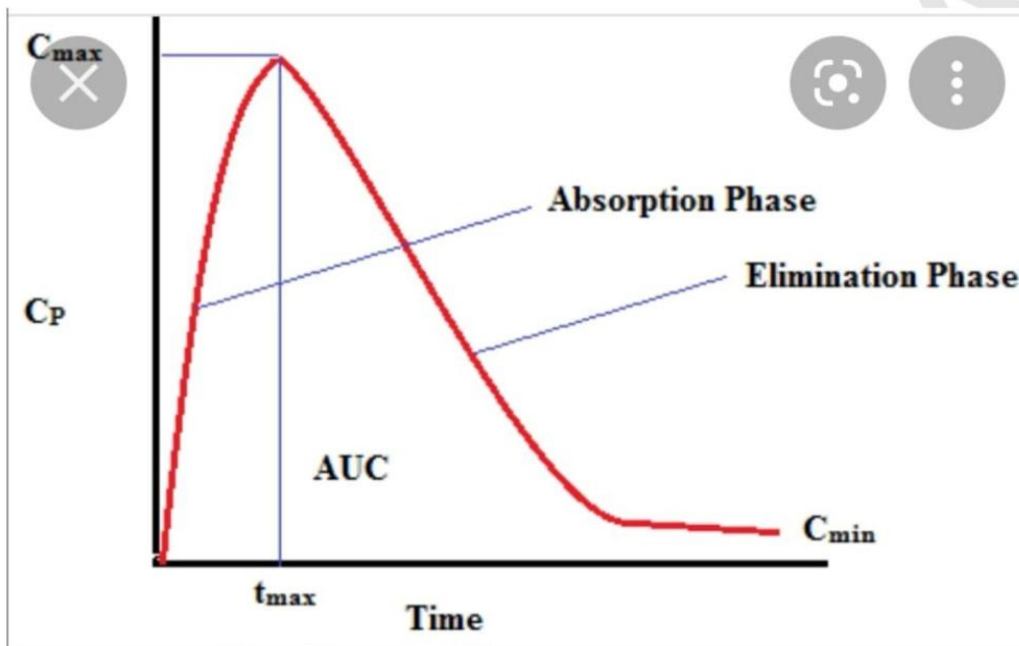
Iv bolus



Iv infusion

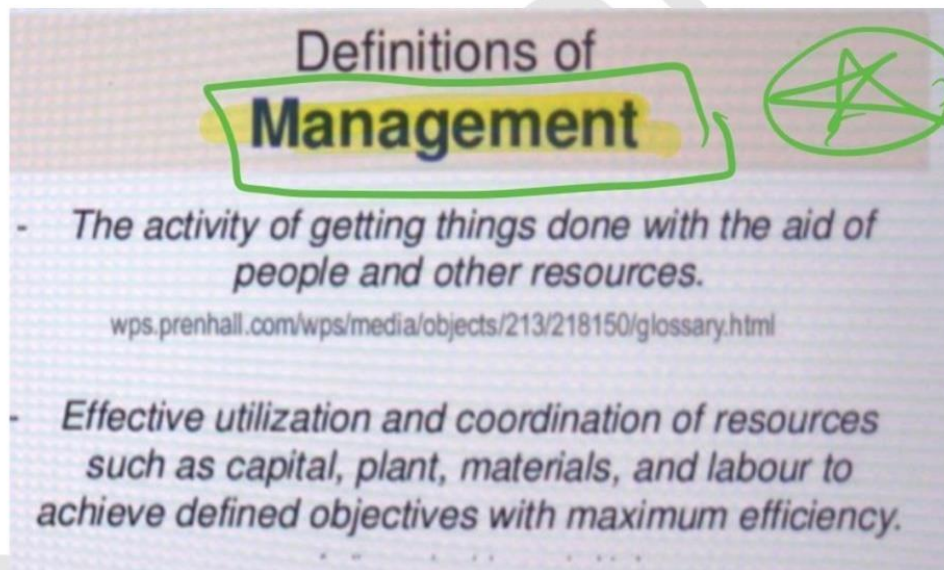


oral



Management

Management is the process of dealing with or controlling things or people. (In general) Management is both an art and science. As a science, it seeks to know variables that affect performance and as an art, it uses the function of coordinating to determine the performance of workers.



Which of the following is the definition of management?

- A) The art of maximizing profitability
- B) The art of maximizing productivity**
- C) The art of maximizing possibilities
- D) The art of maximizing predictability

Pharmacophore

It is part of molecule bind to receptor responsible for activity

Benifience

A person did something good the behavior

bioavailability

Fraction of a dose of drug that is absorbed from its site of administration and reaches, in an unchanged form, the systemic circulation

chemotaxis

Movement of a motile cell or organism, or part of one, in a direction corresponding to a gradient of increasing or decreasing concentration of a particular substance

p value

The p-value is the level of marginal significance within a statistical hypothesis test representing the probability of the occurrence of a given event.

Strategy plan

A systematic process of envisioning a desired future, and translating this vision into broadly defined goals or objectives and a sequence of steps to achieve them.

polypharmacy

is defined as the simultaneous use of multiple drugs to

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treat a single ailment or condition, or the simultaneous use of multiple drugs by a single patient, for one or more conditions. As the population is aging, polypharmacy has become an important risk factor for poor outcomes in the elderly

pharmacovigilance?

the study of the safety of marketed drugs under the practical conditions of clinical use in large communities – involves the paradox that what is probably the most highly regulated industry in the world is, from time to time, forced to remove approved and licensed products from the market because of clinical toxicity.

Autonomy?

Justice ?

Beneficence?

Non-maleficence?

.

(anergy)

in immunity lack of reaction by the body's defense mechanisms to foreign substances

toxidrome ? over poison induced syndrome

tolerence ? patient need more dose to obtain the same effect

bureaucratic

The style of management in which the owner of a pharmacy emphasizes the development of detailed policies and written procedures for employees to observe at all times

Antidote : antagonism of the toxicity of over dose of a

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drug

definition for a soluble solution ? Homogenous

FDA Fast Track what this mean ? drug that shows promising results for life-threatening disease with NO others available can do that

which of the following used to deal with adverse drug effects ? **Pharmacovigilance**

half-life($t_{1/2}$) ?

Time it takes for the plasma concentration or the amount of drug in the body to be reduced by 50%

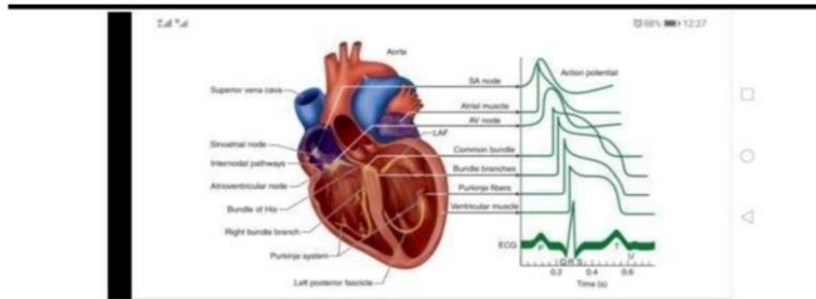
First-Pass Effect Blood that peruses virtually all the gastrointestinal tissues passes through the liver by means of the hepatic portal vein

Bioavailability Area under the curve

Compassionate drug Compassionate drug use (or sometimes just compassionate use) is the use of a new, unapproved drug to treat a seriously ill patients when no other treatments are available

Selection Bias?: proper randomization is not achieved, the study in cleveland is not representative of the US This type of error occurs when a systematic error in the enrollment of individuals in a study causes a biased estimation of the association between exposure and outcome

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Three questions about this picture

2/PR interval represent what

- A/atrial conductivity
- ☒ B/atrioventricular conduction
- C/ventricular conductivity
- D/ventricular plateau (other wards I forgot)

93/ST segment represent what

- A/atrial conductivity
- B/atrioventricular conduction
- ☒ C/ventricular conductivity
- D/ventricular plateau (other wards I forgot)

4/QT represent what

- A/atrial conductivity
- B/atrioventricular conduction
- ☒ C/ventricular conductivity
- D/ventricular plateau (other wards I forgot)

5/pharmacist manager screamed on one of the pharmacist who dispensed wrong medication in front of all the staff what is the good response?

- A/ he should punish him in separate area
- B/ask for a meeting with the staff
- ☒ C/ cut from his salary
- D/ try to find the cause of the error and set solutions

Correct answer ????

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Child-Pugh score

Parameter	1 point	2 points	3 points
ascites	none	slight	moderate to severe
encephalopathy	none	slight to moderate	moderate to severe
total bilirubin (mg/dL)	<2	2-3	>3
albumin (g/dL)	>3,5	2,8-3,5	<2,8
protrombin time (s)	1-3	4-6	>6

Class	Points
Child-Pugh A	5-6
Child-Pugh B	7-9
Child-Pugh C	10-15

For liver disease

Herbal :

5Ht3 ondansetron like : ginger anti emetic

Cough: eucalyptus oil

Expectorant : guaifenesin

UTI inflections: cranberry

Gout: cholechicin

Curarine: muscle relaxant

Active ingredient in plant leaves: chlorophyll

Antioxidants: Grape seed

Painful of menstruation: Black cohosh Contain high concentration of estrogen

Benign Prostatic Hyperplasia BPH : Saw palmetto

Immunostimulants :Echinacea

Anxiety and Depressant: kava kava valerian - St. John's Wort,,lavander

Prevent Cardiovascular disease :Fish oil: Omega 3

To teat High cholesterol and HTN: garlic

ف ملف مجمع herbal باقي

Strengths Characteristics of a business which give it advantages over its competitors	Weaknesses Characteristics of a business which make it disadvantageous relative to competitors
Opportunities Elements in a company's external environment that allow it to formulate and implement strategies to increase profitability	Threats Elements in the external environment that could endanger the integrity and profitability of the business

SWOT ANALYSIS

	Helpful to achieving the objective	Harmful to achieving the objective
Internal origin (attributes of the organization)	Strengths	Weaknesses
External origin (attributes of the environment)	Opportunities	Threats

In SWOT analysis T mean ?

- *threat
- *treat
- *title

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A clothing retail store starts off year 1 with 25 million in the beginning inventory, and throughout year 1, the retailer purchases 10 million in additional inventory and fails to sell 5 million in inventory, ?

- A. 30 million
- B. 20 million
- C. 10 million
- D. 50 Million

Cost of goods sold (COGS)

= begging inventory +purchases in current period - ending inventory

Inventory turnover ?

5		
6	Opening Inventory	3000000
7	Closing Inventory	4000000.00
8	cost of goods sold	1,000,000
9		
10	Average Inventories can be Calculated as:	
11	Average Inventories = (Beginning Inventories + Ending Inventories) / 2	
12		
13	Average Inventories	3500000
14		
15	Inventory Turnover Ratio can be calculated as:	
16	Inventory Turnover Ratio = Cost of Goods Sold/ Average Inventory	
17		
18	Inventory Turnover Ratio Formula	=B8/B13
19	Inventory Turnover Ratio	0.29
20		



<u>LEVELS</u>	
Below basic (level 1)	be able to interpret short, simple text to perform routine tasks. has trouble matching information or identifying numbers to use in mathematical problems.
Basic (level 2)	can solve routine mathematical problems or make simple inferences. However, people with level 1 or level 2 Skills would find it difficult to interpret a dose chart on an OTC cold medication to calculate the correct dose for a child
Intermediate (level 3)	can summarize text, find and apply facts from denser text, and identify and apply information to solve arithmetic calculations.
Proficient (level 4)	can analyze and integrate several pieces of information or solve more abstract or multistep mathematical problems

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An older adult male patient correctly completed the calculation in the Newest Vital Sign (NVS) questionnaire. In the package insert dispensed with his prescriptions, he was able to find relevant facts about adverse effects associated with medications he was taking. He is selecting a Medicare D program and is struggling to compare brochures from several companies to identify the best program for him on the basis of his medications. Which National Assessment of Adult Literacy (NAAL) classification is this patient most likely in?

- A. Below basic
- B. Basic
- C. Intermediate
- D. Proficient

A patient is working with the pharmacist to determine a correct dosing regimen for bolus insulin. The patient has had difficulty calculating the dose given the carbohydrate ratio or correction factor, so the pharmacist created a dosing chart with ranges for the patient's correction factor (e.g., 150—200 mg/dL = 5 units). After a few months, the patient still cannot apply the simplified dosing regimen. Which most closely resembles this patient's National Assessment of Adult Literacy category?

- A. Below basic (level 1).
- B. Basic (level 2).
- C. Intermediate (level 3).
- D. Proficient (level 4).

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المشاركين في الكورس وتعتبر أمانه